

Copy an Entire GNU/Linux Installation and Make it Work



For various reasons, one may need to transfer an entire OS to a different system or an SSD. This seemingly difficult task has been explained in a stepwise manner in this tutorial.

Unless your first computer is still scratch-free and you've kept its warranty card, you probably need to do some migration. Maybe you've got a new machine, or you're eager to know how fast your system boots with an SSD. Even if you don't care about any fancy upgrades, you're left with no other choice if your hard disk is dying.

Hand picking your personal files and copying them is tiresome enough. But you're not yet done. You have to download and install the OS as well as hundreds of packages, and configure each one. If you are a developer, one can bet you'll be spending more time doing these tasks than doing actual development.

That's why everybody wonders if they can *copy an entire system*. Not just some personal files, but the whole OS, the boot setup and the applications.

Upon the next startup, you want to be able to just select the new disk, and what boots has the same old spirit but new flesh.

Sounds good, right? But is it possible? It is, if you are using GNU/Linux and you're not opting for an entirely different hardware architecture. It might be possible with many other operating systems also, excluding the proprietary ones that the developers usually put a lock into.

In this article, our discussion is limited to GNU/Linux, but applies to any distro that hasn't dared to change the very fundamental things.

Methods

There are different methods to get an OS installation cloned into another disk or partition. They include:

- Taking a bit-by-bit copy of the entire disk or a partition (disk